

A
Project Report
On

**“Real-Time Gesture-to-Text Glove for
Speech-Impaired Individuals”**

Submitted in partial fulfillment for the award of the degree of Bachelor of Technology

In

ELECTRONICS ENGINEERING



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DECLARATION

I hereby declare that the work, being presented in the project report entitled “Real-Time Gesture-to-Text Glove for Speech-Impaired Individuals” in partial fulfillment of the requirement for the award of the Degree in Bachelor of Technology in Electronics engineering submitted to the Department of Electronics Engineering of J.C. Bose University of Science and Technology, YMCA, Faridabad is an authentic record of my work carried out during a period from July 2025 to Jan 2025 under the supervision of Dr. Rohit Tripathi, Department of Electronics Engineering. No part of the matter embodied in the project has been submitted to any other University / Institute for the award of any Degree or Diploma.

Abstract

The IoT Glove Project aims to design and develop a smart wearable device capable of detecting and transmitting hand and finger movements using embedded sensors and wireless technology. The glove is equipped with **flex sensors** to measure the degree of finger bending, which are interfaced with an **ESP32 microcontroller**. The microcontroller processes the analog signals from the sensors, converts them into digital data, and transmits the information wirelessly through Wi-Fi or Bluetooth to a remote system or application.

The system effectively transforms hand gestures into digital data that can be used for various applications such as gesture-controlled robotics, prosthetic hand control, sign language interpretation, and rehabilitation monitoring. This project demonstrates the practical use of **IoT, sensor interfacing, and embedded systems** in developing low-cost, portable, and intelligent human-machine interfaces. The implementation highlights the importance of integrating hardware and software to achieve real-time gesture recognition and control.

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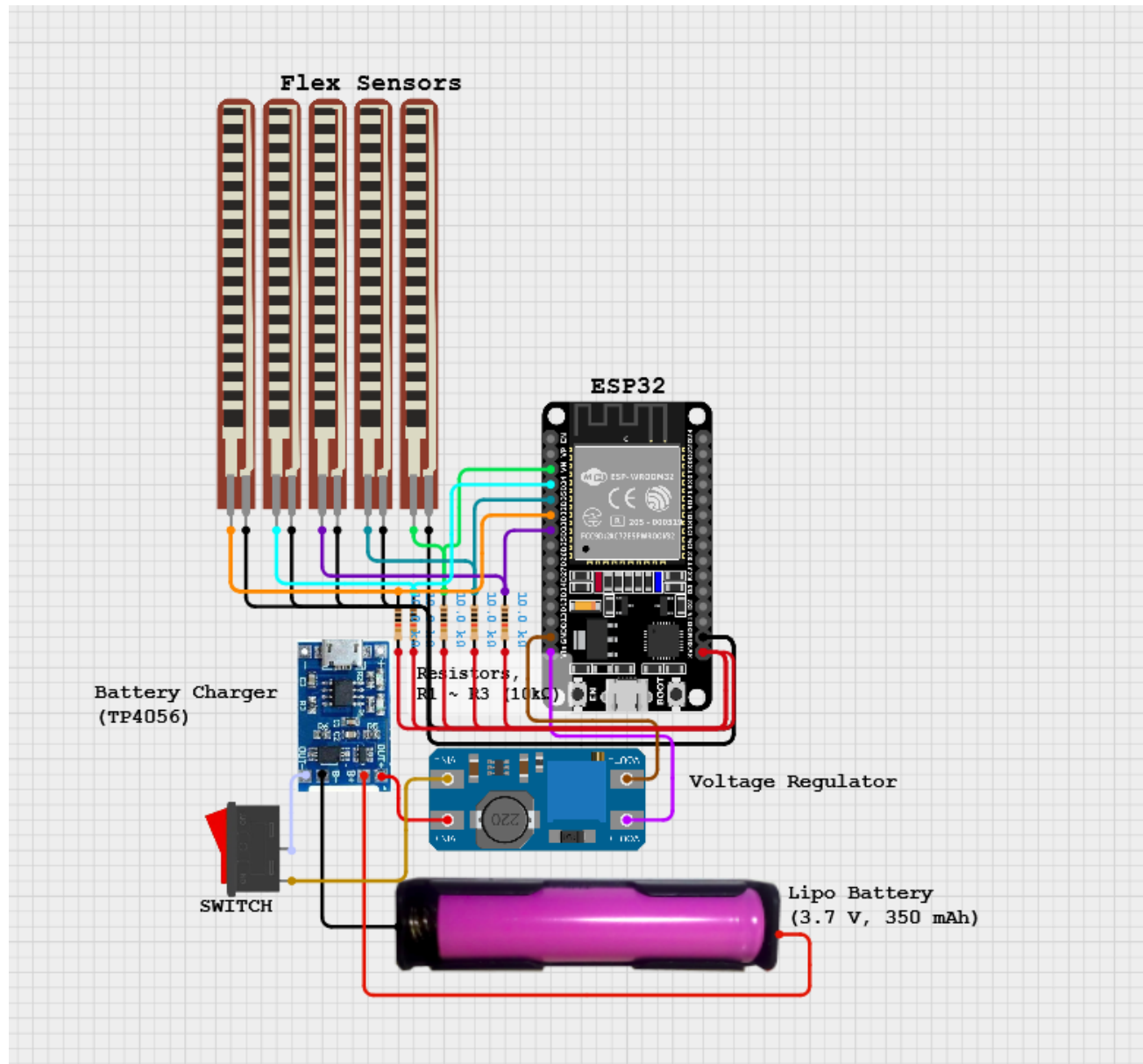
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COMPONENTS USED

The following components for Digital Trainer Kit are required in the project:-

1. ESP32 Microcontroller
2. Flex Sensors
3. Resistors
4. Jumper Wires
5. Breadboard / PCB
6. Li-ion / Li-Po Battery
7. TP4056 Charging Module
8. LEDs
9. Glove (cotton or synthetic)
10. Heat Shrink Tubes

CIRCUIT DIAGRAM



Working Principle

The IoT Glove Project works on the principle of converting hand and finger movements into electrical signals that can be processed and interpreted as meaningful gestures. The glove is embedded with multiple flex sensors, each attached to a finger. These sensors vary their resistance depending on the degree of bending of the finger — the more the finger bends, the greater the change in resistance. This change in resistance is converted into a corresponding voltage signal using voltage divider circuits. The ESP32 microcontroller reads these analog voltage values through its ADC (Analog-to-Digital Converter) pins and processes them to determine the position of each finger. Based on the combination of finger bends, the ESP32 identifies specific gestures by comparing the sensor values with predefined threshold ranges stored in its program.

Once a particular gesture is detected, the ESP32 encodes the gesture data and transmits it wirelessly via Bluetooth communication to a paired Android application. The Android app receives this data, decodes it, and displays the corresponding word or sentence associated with that gesture on the screen. This creates a seamless system capable of translating physical hand gestures into readable text in real time. The overall working demonstrates the integration of sensor technology, embedded processing, and wireless communication to build an intelligent, low-cost, and efficient gesture recognition system.

CONCLUSION AND FUTURE Enhancements

The proposed **Real-Time Gesture-to-Text Glove** is an innovative IoT-based assistive technology aimed at empowering speech-impaired individuals. It combines wearable electronics, gesture recognition, and Bluetooth-enabled Android app support to provide an affordable, user-friendly communication tool. This project can greatly improve accessibility, inclusivity, and independence for people with communication challenges.

Future Enhancements

- Integration with **TensorFlow Lite Micro** for advanced gesture recognition.
- Multi-language support in the Android app.
- Integration with cloud services for storing and personalizing gesture libraries.
- Adding optional **text-to-speech** functionality inside the mobile app.

Thank You